WORKED SOLUTION

STEP I 2011, Question 11

Mechanics • Equilibrium

STEP I 2011, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

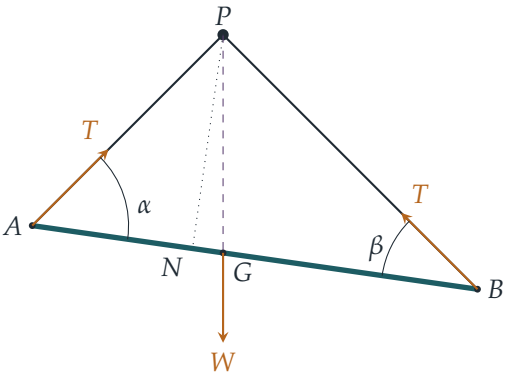
step.maths.org/questions/11-s1-q11

The key idea

Only three forces act on the bar: the two string tensions at A and B, and its weight at G. Because the peg is smooth, the two tensions are equal, so taking moments about G (where the weight has no moment) gives the relation between the angles at once. For the second part, the extra fact we need is that three forces in equilibrium that are not all parallel must pass through one point. Here that point is the peg, so PG is vertical.

Solution

We model the bar as a rigid rod, the string as light and inextensible, and the peg as smooth. Let the weight of the bar be W.



Showing that $3\sin\alpha = 4\sin\beta$

The string is light and passes over a smooth peg, so its tension T is the same throughout. The bar therefore feels a force T at A along AP and a force T at B along BP, as well as W acting vertically at G.

Take moments about G. The weight has no moment. The perpendicular distance from G to the line AP is $AG\sin\alpha = 3d\sin\alpha$, and from G to the line BP it is $GB\sin\beta = 4d\sin\beta$, since $GB = 7d - 3d = 4d$. The tension at A turns the bar one way about G and the tension at B turns it the other way, so for equilibrium

$$T\cdot 3d\sin\alpha = T\cdot 4d\sin\beta \quad\Longrightarrow\quad 3\sin\alpha = 4\sin\beta,$$

as required (dividing by $Td \neq 0$).

The length of the string

With $\cos\beta = \frac{4}{5}$ we have $\sin\beta = \frac{3}{5}$ (β is an angle of a triangle, so $\sin\beta > 0$). Then

$$\sin\alpha = \tfrac{4}{3}\sin\beta = \tfrac{4}{5}, \qquad \cos\alpha = \tfrac{3}{5},$$

taking the positive root because α is acute. Now

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta = \tfrac{3}{5}\cdot\tfrac{4}{5} - \tfrac{4}{5}\cdot\tfrac{3}{5} = 0,$$

and $0 < \alpha+\beta < \pi$, so $\alpha+\beta = \frac{\pi}{2}$. The triangle PAB therefore has a right angle at P, with hypotenuse $AB = 7d$. Hence

$$PA = 7d\cos\alpha = \tfrac{21}{5}d, \qquad PB = 7d\cos\beta = \tfrac{28}{5}d,$$

and the length of the string is

$$PA + PB = \tfrac{49}{5}d.$$

The inclination of the bar

? **Ask yourself** We know where P is relative to the bar, but where is "vertical"?

The two tensions act along lines through P, so their resultant also acts through P. For equilibrium this resultant must balance the weight, so it must be equal and opposite to W and along the same line. (Three non-parallel forces in equilibrium are concurrent.) So the vertical line through G passes through P: *PG is vertical*.

Let N be the foot of the perpendicular from P to AB. In the right-angled triangle PAN,

$$AN = PA\cos\alpha = \tfrac{21}{5}d\cdot\tfrac{3}{5} = \tfrac{63}{25}d, \qquad PN = PA\sin\alpha = \tfrac{21}{5}d\cdot\tfrac{4}{5} = \tfrac{84}{25}d.$$

Since $AN = \frac{63}{25}d < 3d = AG$, the point N lies between A and G, with

$$NG = 3d - \tfrac{63}{25}d = \tfrac{12}{25}d.$$

In the right-angled triangle PNG,

$$\tan\angle PGN = \frac{PN}{NG} = \frac{84}{12} = 7.$$

The angle PGN is the angle between the bar and the vertical GP, measured on the A side. It is acute, so the end A is above the horizontal through G and B is below it, which

matches the given configuration. The angle θ between the bar and the horizontal is $\frac{\pi}{2} - \angle PGN$, so

$$\tan\theta = \frac{1}{\tan\angle PGN} = \frac{1}{7}, \qquad \text{that is,} \qquad \theta = \arctan\tfrac{1}{7},$$

as required.

Checks

- The string, $\frac{49}{5}d = 9.8d$, is longer than the bar, $7d$, as the triangle inequality in PAB demands.
- The resultant of two equal tensions bisects the angle between them, so PG bisects angle APB. By the angle bisector theorem, $PA : PB$ should equal $AG : GB = 3 : 4$, and indeed $\frac{21}{5} : \frac{28}{5} = 3 : 4$.
- As a numerical check, we put P on the ellipse with foci A and B (the set of possible peg positions for this string length) and looked for the inclination at which the depth of G below P is stationary. It came out at 8.13°, which agrees with $\arctan\frac{1}{7} \approx 8.130^\circ$.

Common slips

- Giving the two parts of the string different tensions. A smooth peg cannot change the tension in a light string.
- Swapping the distances and getting $4\sin\alpha = 3\sin\beta$. The longer arm $GB = 4d$ goes with the angle at B.
- Forgetting that α is acute and allowing $\cos\alpha = -\frac{3}{5}$. That gives a different triangle, and in it $\alpha + \beta$ is not a right angle.

Beyond STEP

For a fixed string length, the peg must lie on an ellipse with foci A and B. The fact that PG bisects angle APB is then the reflection property of the ellipse: the normal at P bisects the angle between the focal radii. Equilibria are also exactly the positions where the potential energy is stationary, and this is the starting point of the principle of virtual work and of Lagrangian mechanics in a first-year course. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to read about equilibrium, torque and energy methods.

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